

Standard Model Signal in Cosmological Higgs Collider

1 Gauge Boson Signal

Higgs bulk-boundary propagator is a massless propagator:

$$G_{\pm}(k; \tau) = \frac{H^2}{2k^3} (1 \mp ik\tau) e^{\pm ik\tau} \quad (1)$$

The Higgs-gauged coupling are from the original SM Lagrangian:

$$\frac{\sqrt{-g}}{2} |D_{\mu} \mathbf{H}|^2 \supset \frac{a(\tau)^2}{4} g^2 (h_0 + \delta h)^2 \left(W_{\mu}^+ W^{-\mu} + \frac{1}{2c_W^2} Z^{\mu} Z_{\mu} \right) \quad (2)$$

The masses of gauge bosons due to the infrared Higgs loop contribution are:

$$m_W^2 = \frac{3g^2 H^4}{8\pi^2 m_H^2}, \quad m_Z^2 = \frac{m_W^2}{c_W^2} \quad (3)$$

Mode function of a massive gauge boson [1]:

$$B^{(\pm)}(k, \tau) = \frac{1}{\sqrt{2k}} W_{0, i\tilde{\nu}}(2ik\tau), \quad (4)$$

$$B^{(T)}(k, \tau) = \frac{\sqrt{\pi}k}{2m} e^{-\pi\tilde{\nu}/2} H(-\tau)^{3/2} H_{i\tilde{\nu}}^{(1)}(-k\tau), \quad (5)$$

$$B^{(L)}(k, \tau) = -\frac{i\sqrt{\pi}}{4m} e^{-\pi\tilde{\nu}/2} H(-\tau)^{1/2} \left[H_{i\tilde{\nu}}^{(1)}(-k\tau) - k\tau \left(H_{1+i\tilde{\nu}}^{(1)}(-k\tau) - H_{-1+i\tilde{\nu}}^{(1)}(-k\tau) \right) \right] \quad (6)$$

where $\tilde{\nu} = \sqrt{(m/H)^2 - 1/4}$. And the W function with zero chemical potential can transform to a Hankel function:

$$\begin{aligned} W_{0, i\tilde{\nu}}(2ik\tau) &= \sqrt{\frac{2ik\tau}{\pi}} K_{i\tilde{\nu}}(ik\tau) \\ &= \sqrt{\frac{i\pi k\tau}{2}} i e^{-\pi\tilde{\nu}/2} H_{i\tilde{\nu}}^{(1)}(-k\tau) \end{aligned} \quad (7)$$

So the grater propagator or $+-$ propagator of each polarization:

$$D_{>}^{(\pm)}(k; \tau_1, \tau_2) = B^{(\pm)}(k, \tau_1) B^{(\pm)*}(k, \tau_2) = \frac{\pi e^{-\pi\tilde{\nu}} (\tau_1 \tau_2)^{1/2}}{4} H_{i\tilde{\nu}}^{(1)}(-k\tau_1) H_{-i\tilde{\nu}}^{(2)}(-k\tau_2), \quad (8)$$

$$D_{>}^{(T)}(k; \tau_1, \tau_2) = B^{(T)}(k, \tau_1) B^{(T)*}(k, \tau_2) = \frac{\pi H^2 k^2}{4m^2} e^{-\pi\tilde{\nu}} (\tau_1 \tau_2)^{3/2} H_{i\tilde{\nu}}^{(1)}(-k\tau_1) H_{-i\tilde{\nu}}^{(2)}(-k\tau_2), \quad (9)$$

$$D_{>}^{(L)}(k; \tau_1, \tau_2) = \frac{1}{k^2} \left(\partial_{\tau_1} - \frac{2}{\tau_1} \right) \left(\partial_{\tau_2} - \frac{2}{\tau_2} \right) D_{>}^{(T)}(k; \tau_1, \tau_2). \quad (10)$$

We quote the scalar propagator with mass index $i\tilde{\nu}$:

$$D_{>}^{(\text{scalar})} = \frac{\pi}{4} e^{-\pi\tilde{\nu}} (\tau_1\tau_2)^{3/2} H_{i\tilde{\nu}}^{(1)}(-k\tau_1) H_{-i\tilde{\nu}}^{(2)}(-k\tau_2) \quad (11)$$

and the nonlocal part of its late time expansion:

$$[D_{>}]_{\text{NonL}} = [D_{<}]_{\text{NonL}} = \frac{(\tau_1\tau_2)^{3/2}}{4\pi} \left[\Gamma^2(-i\tilde{\nu}) \left(\frac{k^2\tau_1\tau_2}{4} \right)^{i\tilde{\nu}} + \text{c.c.} \right] \quad (12)$$

And the vector propagator can be related to the scalar propagator:

$$D_{>}^{(\pm)} = \frac{1}{\tau_1\tau_2} D_{>}^{(\text{scalar})} \quad (13)$$

$$D_{>}^{(T)} = \frac{k^2}{\tilde{\nu}^2 + 1/4} D_{>}^{(\text{scalar})} \quad (14)$$

$$D_{>}^{(L)} = \frac{1}{\tilde{\nu}^2 + 1/4} \left(\partial_{\tau_1} - \frac{2}{\tau_1} \right) \left(\partial_{\tau_2} - \frac{2}{\tau_2} \right) D_{>}^{(\text{scalar})}(k; \tau_1, \tau_2) \quad (15)$$

We can see the propagator of temporal component of longitudinal mode have a suppression factor $(k\tau)^2$, so it will contribute the next leading term and we will ignore it below.

And the nonlocal part:

$$[D^{(\pm)}]_{\text{NonL}} = \frac{(\tau_1\tau_2)^{1/2}}{4\pi} \left[\Gamma^2(-i\tilde{\nu}) \left(\frac{k^2\tau_1\tau_2}{4} \right)^{i\tilde{\nu}} + \text{c.c.} \right] \quad (16)$$

$$[D^{(T)}]_{\text{NonL}} = \frac{k^2}{\tilde{\nu}^2 + 1/4} \frac{(\tau_1\tau_2)^{3/2}}{4\pi} \left[\Gamma^2(-i\tilde{\nu}) \left(\frac{k^2\tau_1\tau_2}{4} \right)^{i\tilde{\nu}} + \text{c.c.} \right] \quad (17)$$

$$[D^{(L)}]_{\text{NonL}} = \frac{(-1/2 + i\tilde{\nu})^2 (\tau_1\tau_2)^{1/2}}{\tilde{\nu}^2 + 1/4} \frac{1}{4\pi} \left[\Gamma^2(-i\tilde{\nu}) \left(\frac{k^2\tau_1\tau_2}{4} \right)^{i\tilde{\nu}} + \text{c.c.} \right] \quad (18)$$

We also define the propagator of helicity eigenstates:

$$D_{>\mu\nu}^{(\pm 1)}(\mathbf{k}; \tau_1, \tau_2) = e_{\mu,\mathbf{k}}^{(\pm)} e_{\nu,\mathbf{k}}^{(\pm)*} D_{>}^{\pm}(k; \tau_1, \tau_2) \quad (19)$$

$$D_{>\mu\nu}^{(0)}(\mathbf{k}; \tau_1, \tau_2) = e_{\mu,\mathbf{k}}^{(L)} e_{\nu,\mathbf{k}}^{(L)*} D_{>}^{(L)}(k; \tau_1, \tau_2) \quad (20)$$

$$D_{>\mu\nu}(\mathbf{k}; \tau_1, \tau_2) = \sum_{h=\pm 1, 0} D_{>\mu\nu}^{(h)}(k; \tau_1, \tau_2) \quad (21)$$

For $\mathbf{k} = (0, 0, k)$, the polarization vector is:

$$e_{\mu,\mathbf{k}}^{(T)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad e_{\mu,\mathbf{k}}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix}, \quad e_{\mu,\mathbf{k}}^{(L)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad k_{\mu} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ k \end{pmatrix}, \quad (22)$$

and when we rotate $\mathbf{k}$, the $e_{\mu,\mathbf{k}}^{(T)}$ is invariant and the other two types of polarization vector will transform $e'_{i,\mathbf{k}} = R_{ij} e_{j,\mathbf{k}}$:

$$e_{\mu,\mathbf{k}}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \cos\theta \cos\phi \mp i \sin\phi \\ \cos\theta \sin\phi \pm i \cos\phi \\ -\sin\theta \end{pmatrix}, \quad k_{\mu} = e_{\mu,\mathbf{k}}^{(L)} = \begin{pmatrix} 0 \\ \sin\theta \cos\phi \\ \sin\theta \sin\phi \\ \cos\theta \end{pmatrix} \quad (23)$$

The MB representation of the scalar propagator:

$$D_{\geq}(k; \tau_1, \tau_2) = \frac{1}{4\pi} \int_{-\infty}^{+\infty} \frac{ds_1}{2\pi i} \frac{ds_2}{2\pi i} e^{\pm i\pi(s_1-s_2)} \left(\frac{k}{2}\right)^{-2s_{12}} (-\tau_1)^{-2s_1+3/2} (-\tau_2)^{-2s_2+3/2} \quad (24)$$

$$\times \Gamma\left[s_1 - \frac{i\tilde{\nu}}{2}, s_1 + \frac{i\tilde{\nu}}{2}, s_2 - \frac{i\tilde{\nu}}{2}, s_2 + \frac{i\tilde{\nu}}{2}\right] \quad (25)$$

For simplicity we define some loop seeds:

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} |\mathbf{q}|^{-2a} |\mathbf{q} + \mathbf{k}_s|^{-2b} = \frac{k_s^{3-2a-2b}}{(4\pi)^{3/2}} \cdot f_0(a, b) \quad (26)$$

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} |\mathbf{q}|^{-2a} |\mathbf{q} + \mathbf{k}_s|^{-2b} \cos\theta = \frac{k_s^{3-2a-2b}}{(4\pi)^{3/2}} \cdot f_c(a, b) \quad (27)$$

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} |\mathbf{q}|^{-2a} |\mathbf{q} + \mathbf{k}_s|^{-2b} \cos^2\theta = \frac{k_s^{3-2a-2b}}{(4\pi)^{3/2}} \cdot f_{cc}(a, b) \quad (28)$$

where $f(a, b)$ are some Gamma functions arising from the loop integral:

$$f_0(a, b) = \Gamma\left[\begin{matrix} a+b-\frac{3}{2}, \frac{3}{2}-a, \frac{3}{2}-b \\ 3-a-b, a, b \end{matrix}\right] \quad (29)$$

$$f_c(a, b) = \frac{1}{2}f_0(a+1/2, b-1) - \frac{1}{2}f_0(a-1/2, b) - \frac{1}{2}f_0(a+1/2, b) \quad (30)$$

$$f_{cc}(a, b) = \frac{1}{4}f_0(a+1, b-2) + \frac{1}{4}f_0(a-1, b) + \frac{1}{4}f_0(a+1, b) - \frac{1}{2}f_0(a, b-1) - \frac{1}{2}f_0(a+1, b-1) + \frac{1}{2}f_0(a, b) \quad (31)$$

2 Bubble Diagram

For the Bubble diagram:

$$\begin{aligned} \langle \delta h^4 \rangle^{\text{Bubble}} &= (ig^2)^2 \sum_{a,b=\pm} (ab) \int \frac{d\tau_1}{|H\tau_1|^2} \frac{d\tau_2}{|H\tau_2|^2} G_a(k_1; \tau_1) G_a(k_2; \tau_1) G_b(k_3; \tau_2) G_b(k_4; \tau_2) \\ &\times \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{\mu\nu}(q, \tau_1, \tau_2)]_{ab} [D^{\nu\mu}(|\mathbf{k}_s + \mathbf{q}|, \tau_2, \tau_1)]_{ba} \end{aligned} \quad (32)$$

The loop integral part is:

$$\mathcal{LOOP}_{ab} = \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{\mu\nu}(q, \tau_1, \tau_2)]_{ab} [D^{\nu\mu}(|\mathbf{k}_s + \mathbf{q}|, \tau_2, \tau_1)]_{ba} \quad (33)$$

Using the cutting rule, we know that the nonlocal signal part is independent of the SK index:

$$\begin{aligned} \mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(L)i} e_{\mathbf{q}}^{(L)j*} + D^{(\pm)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(\pm)i} e_{\mathbf{q}}^{(\pm)j*} \right] \\ &\times \left[D^{(L)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(L)j} e_{\mathbf{K}}^{(L)i*} + D^{(\pm)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(\pm)j} e_{\mathbf{K}}^{(\pm)i*} \right] \end{aligned} \quad (34)$$

$$= \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{LOOP}_{\sigma_1\sigma_2} \quad (35)$$

where $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$. So we should deal with different polarization mixing. We set $\mathbf{k}_s = k_s(0, 0, 1)$ and $\mathbf{q} = q(\sin \theta \sin \phi, \sin \theta \cos \phi, \cos \theta)$, then the polarization vector become:

$$e_{i,\mathbf{q}}^{(L)} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}, \quad e_{i,\mathbf{q}}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} \cos \theta \cos \phi \mp i \sin \phi \\ \cos \theta \sin \phi \pm i \cos \phi \\ -\sin \theta \end{pmatrix} \quad (36)$$

$$\mathbf{e}_{i,\mathbf{K}}^{(L)} = \frac{1}{K} \begin{pmatrix} q \sin \theta \cos \phi \\ q \sin \theta \sin \phi \\ q \cos \theta + k_s \end{pmatrix}, \quad \mathbf{e}_{i,\mathbf{K}}^{(\pm)} = \frac{1}{\sqrt{2}K} \begin{pmatrix} \cos \phi (q \cos \theta + k_s) \mp i \sin \phi K \\ \sin \phi (q \cos \theta + k_s) \pm i \cos \phi K \\ -q \sin \theta \end{pmatrix} \quad (37)$$

The signal part is contributed from the loop integral of nonlocal part of each propagator. And we calculate the first term as an example:

$$\begin{aligned} \mathcal{LOOP}_{LL} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(L)i} e_{\mathbf{q}}^{(L)j*} \cdot D^{(L)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(L)j} e_{\mathbf{K}}^{(L)i*} \right]_{\text{NonL}} \\ &= \sum_{c=\pm} \frac{(-1/2 + i\tilde{\nu})^4 (\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(\tilde{\nu}^2 + 1/4)^2 (4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[\left(\frac{q}{2}\right)^{2ic\tilde{\nu}} \left(\frac{K}{2}\right)^{2ic\tilde{\nu}} |e_{\mathbf{q}}^{(L)} \cdot e_{\mathbf{K}}^{(L)*}|^2 \right] \end{aligned} \quad (38)$$

Completing the loop integral:

$$\begin{aligned} \mathcal{LOOP}_{LL} &= \sum_{c=\pm} \frac{(-1/2 + i\tilde{\nu})^4 (\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(\tilde{\nu}^2 + 1/4)^2 (4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \cdot \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\quad \times (f_0(x-1, x+1) + 2f_c(x-1/2, x+1) + f_{cc}(x, x+1)) \end{aligned} \quad (39)$$

where $x = -ic\tilde{\nu}$. Using the identity $\mathbf{q} \cdot \mathbf{K} = \frac{1}{2}(q^2 + K^2 - k_s^2)$, we can also express it as:

$$\begin{aligned} \mathcal{LOOP}_{LL} &= \sum_{c=\pm} \frac{(-1/2 + i\tilde{\nu})^4 (\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(\tilde{\nu}^2 + 1/4)^2 (4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \cdot \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\quad \times \frac{1}{4} (f(x-1, x+1) + f(x+1, x-1) + f(x+1, x+1) \\ &\quad + 2f(x, x) - 2f(x, x+1) - 2f(x+1, x)) \end{aligned} \quad (40)$$

We denote it as:

$$\mathcal{LOOP}_{LL} = \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \mathcal{I}_{LL}(-ic\tilde{\nu}) \quad (41)$$

And also for the transversal mode:

$$\begin{aligned} \mathcal{LOOP}_{++} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(+)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(+i)} e_{\mathbf{q}}^{(+j)*} \cdot D^{(+)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(+j)} e_{\mathbf{K}}^{(+i)*} \right]_{\text{NonL}} \\ &= \sum_{c=\pm} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[\left(\frac{q}{2}\right)^{2ic\tilde{\nu}} \left(\frac{K}{2}\right)^{2ic\tilde{\nu}} |e_{\mathbf{q}}^{(+)} \cdot e_{\mathbf{K}}^{(+)*}|^2 \right] \end{aligned} \quad (42)$$

Also considering the vector structure:

$$\begin{aligned}\mathcal{LOOP}_{++} &= \sum_{c=\pm} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \times \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\times \frac{1}{4} [f_0(x, x) + f_0(x-1, x+1) + f_{cc}(x, x+1) \\ &+ 2f_0(x - \frac{1}{2}, x + \frac{1}{2}) + 2f_c(x - \frac{1}{2}, x+1) + 2f_c(x, x + \frac{1}{2})] \end{aligned} \quad (43)$$

$$= \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \mathcal{I}_{++}(-ic\tilde{\nu}) \quad (44)$$

We note that the $--$ term equals to the $++$ term. Then we should deal with the mixing term. Firstly we consider the $L+$ term:

$$\begin{aligned}\mathcal{LOOP}_{L+} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(+)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(+i)} e_{\mathbf{q}}^{(+j)*} \cdot D^{(+)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(+j)} e_{\mathbf{K}}^{(+i)*} \right]_{\text{NonL}} \\ &= \sum_{c=\pm} \frac{(-1/2 + i\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[\left(\frac{q}{2}\right)^{2ic\tilde{\nu}} \left(\frac{K}{2}\right)^{2ic\tilde{\nu}} |e_{\mathbf{q}}^{(L)} \cdot e_{\mathbf{K}}^{(+)*}|^2 \right] \end{aligned} \quad (45)$$

and:

$$\begin{aligned}\mathcal{LOOP}_{L+} &= \sum_{c=\pm} \frac{(-1/2 + i\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \cdot \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\times \frac{1}{2} (f_0(x, x+1) - f_{cc}(x, x+1)) \end{aligned} \quad (46)$$

$$= \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \mathcal{I}_{L+}(-ic\tilde{\nu}) \quad (47)$$

and by completing the dot product we note that $L- = L+ = -L = +L$.

And for $+-$ term:

$$\begin{aligned}\mathcal{LOOP}_{+-} &= \sum_{c=\pm} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \times \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\times \frac{1}{4} [f_0(x, x) + f_0(x-1, x+1) + f_{cc}(x, x+1) \\ &- 2f_0(x - \frac{1}{2}, x + \frac{1}{2}) + 2f_c(x - \frac{1}{2}, x+1) - 2f_c(x, x + \frac{1}{2})] \end{aligned} \quad (48)$$

$$= \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \mathcal{I}_{+-}(-ic\tilde{\nu}) \quad (49)$$

and we have $-+ = +-.$

So the loop integral:

$$\mathcal{LOOP} = \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{LOOP}_{\sigma_1 \sigma_2} = \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \mathcal{B}_c \quad (50)$$

where the bubble subgraph contribution is:

$$\mathcal{B}_c = \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \sum_{\lambda_1, \lambda_2=L, \pm} \mathcal{I}_{\lambda_1 \lambda_2}(-ic\tilde{\nu}) \quad (51)$$

Then we can consider the left and right subgraph contribution, namely the time integral. For the left subgraph:

$$\begin{aligned}
\mathcal{T}_c^{(L)} &= (ig^2) \sum_a a \int \frac{d\tau_1}{|H\tau_1|^2} G_a(k_1; \tau_1) G_a(k_2; \tau_1) (-\tau_1)^{1+2ic\tilde{\nu}} \\
&= (ig^2) \frac{H^2}{4k_1^3 k_2^3} \sum_{a=\pm} a \int_{-\infty}^0 d\tau_1 (-\tau_1)^{-1+2ic\tilde{\nu}} (1 - iak_{12}\tau_1 - k_1 k_2 \tau_1^2) e^{iak_{12}\tau_1} \\
&= (ig^2) \frac{H^2}{4k_1^3 k_2^3} \times k_{12}^{-2ic\tilde{\nu}} \cdot \mathcal{P}_c^{(1)}(\tilde{\nu})
\end{aligned} \tag{52}$$

where the coefficient is:

$$\mathcal{P}_c^{(1)}(\tilde{\nu}) = \frac{2 + ic\tilde{\nu}}{2ic\tilde{\nu}} \sinh(\pi c\tilde{\nu}) \Gamma(2 + 2ic\tilde{\nu}) \tag{53}$$

Also for the right subgraph:

$$\mathcal{T}_c^{(R)} = (ig^2) \frac{H^2}{4k_3^3 k_4^3} \times k_{34}^{-2ic\tilde{\nu}} \cdot \mathcal{P}_c^{(1)}(\tilde{\nu}) \tag{54}$$

So the total bubble graph contribution:

$$\langle \delta h^4 \rangle^{\text{Bubble}} = \sum_{c=\pm} \mathcal{T}_c^{(L)} \mathcal{T}_c^{(R)} \mathcal{B}_c \tag{55}$$

$$= (ig^2)^2 \frac{H^4 k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times C^{\text{Bub}}(\tilde{\nu}) + \text{c.c.} \tag{56}$$

where the magnitude is:

$$C^{\text{Bub}}(\tilde{\nu}) = [\mathcal{P}_+^{(1)}(\tilde{\nu})]^2 \cdot \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \cdot \sum_{\lambda_1, \lambda_2=L, \pm} \mathcal{I}_{\lambda_1 \lambda_2}(\tilde{\nu})|_{c=+} \tag{57}$$

3 Triangle Diagram

For one of the triangle diagrams:

$$\begin{aligned}
\langle \delta h^4 \rangle^{\text{Tri1}} &= (ig^2) (ig^2 h_0)^2 \sum_{a,b,c=\pm} (abc) \int \frac{d\tau_1}{|H\tau_1|^2} \frac{d\tau_2}{|H\tau_2|^2} \frac{d\tau_3}{|H\tau_3|^2} G_a(k_1; \tau_1) G_a(k_2; \tau_2) G_c(k_3; \tau_3) G_c(k_4; \tau_3) \\
&\times \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)]_{bc} [D_{ki}(\mathbf{q}, \tau_3, \tau_1)]_{ca}
\end{aligned} \tag{58}$$

Firstly we deal with the loop integral:

$$\mathcal{LOOP} = \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)]_{bc} [D_{ki}(\mathbf{q}, \tau_3, \tau_1)]_{ca} \tag{59}$$

Using the cutting rule, we know that only when $\mathbf{q}$ and $\mathbf{k}_s$ both become soft, this integral will contribute the nonlocal signal.

$$\mathcal{LOOP} = \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)] [D_{ki}(\mathbf{q}, \tau_3, \tau_1)] \tag{60}$$

Here we use:

$$e_{i,\mathbf{k}_1}^{(L)} = \begin{pmatrix} \sin \theta_1 \\ 0 \\ \cos \theta_1 \end{pmatrix}, \quad e_{i,\mathbf{k}_1}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} \cos \theta_1 \\ \pm i \\ -\sin \theta_1 \end{pmatrix}. \quad (61)$$

and expanding the propagator by helicity basis:

$$\begin{aligned} \mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(k_1, \tau_1, \tau_2) e_{\mathbf{k}_1}^{(L)i} e_{\mathbf{k}_1}^{(L)j*} + D^{(\pm)}(k_1, \tau_1, \tau_2) e_{\mathbf{k}_1}^{(\pm)i} e_{\mathbf{k}_1}^{(\pm)j*} \right]_{ab} \\ &\quad \times \left[D^{(L)}(K, \tau_2, \tau_3) e_{\mathbf{K}}^{(L)j} e_{\mathbf{K}}^{(L)k*} + D^{(\pm)}(K, \tau_2, \tau_3) e_{\mathbf{K}}^{(\pm)j} e_{\mathbf{K}}^{(\pm)k*} \right] \\ &\quad \times \left[D^{(L)}(q, \tau_3, \tau_1) e_{\mathbf{q}}^{(L)k} e_{\mathbf{q}}^{(L)i*} + D^{(\pm)}(q, \tau_3, \tau_1) e_{\mathbf{q}}^{(\pm)k} e_{\mathbf{q}}^{(\pm)i*} \right] \end{aligned} \quad (62)$$

$$= \sum_{\lambda, \sigma_1, \sigma_2 = L, \pm} \mathcal{LOOP}_{\sigma_1 \sigma_2}^\lambda \quad (63)$$

where each porlarization combination term is:

$$\begin{aligned} \mathcal{LOOP}_{\sigma_1 \sigma_2}^\lambda &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D^{(\lambda)}(k_1, \tau_1, \tau_2)]_{ab} D^{(\sigma_1)}(K, \tau_2, \tau_3) D^{(\sigma_2)}(q, \tau_3, \tau_1) \\ &\quad \times e_{\mathbf{k}_1}^{(\lambda)i} e_{\mathbf{k}_1}^{(\lambda)j*} e_{\mathbf{K}}^{(\sigma_1)j} e_{\mathbf{K}}^{(\sigma_1)k*} e_{\mathbf{q}}^{(\sigma_2)k} e_{\mathbf{q}}^{(\sigma_2)i*} \end{aligned} \quad (64)$$

Firstly we consider the $\lambda = L$ terms and complete the contraction with the other two propagators. We set $\mathbf{k}_1 = k_1(\sin \theta_1, 0, \cos \theta_1)$, and still use $\mathbf{k}_s = k_s(0, 0, 1)$ and $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$. Then the vector structure is:

$$\begin{aligned} \mathcal{E}_{\sigma_1 \sigma_2}^L &= e_{\mathbf{k}_1}^{(L)i} e_{\mathbf{k}_1}^{(L)j*} e_{\mathbf{K}}^{(\sigma_1)j} e_{\mathbf{K}}^{(\sigma_1)k*} e_{\mathbf{q}}^{(\sigma_2)k} e_{\mathbf{q}}^{(\sigma_2)i*} = (e_{\mathbf{k}_1}^{(L)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}) (e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}) (e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(L)}) \\ &= A_{L\sigma_1} B_{\sigma_1 \sigma_2} C_{\sigma_2 L} \end{aligned} \quad (65)$$

where $\sigma_1, \sigma_2 = L, \pm$. For simplicity we define:

$$A_{LL} = \frac{q \sin \theta \sin \theta_1 \cos \phi + (q \cos \theta + k_s) \cos \theta_1}{K}, \quad (66)$$

$$A_{L+} = \frac{\sin \theta_1 \cos \phi (q \cos \theta + k_s) - iK \sin \theta_1 \sin \phi - q \sin \theta \cos \theta_1}{\sqrt{2}K}, \quad (67)$$

$$A_{L-} = \frac{\sin \theta_1 \cos \phi (q \cos \theta + k_s) + iK \sin \theta_1 \sin \phi - q \sin \theta \cos \theta_1}{\sqrt{2}K}, \quad (68)$$

$$C_{LL} = \sin \theta \sin \theta_1 \cos \phi + \cos \theta \cos \theta_1, \quad (69)$$

$$C_{+L} = \frac{\sin \theta_1 (\cos \theta \cos \phi + i \sin \phi) - \sin \theta \cos \theta_1}{\sqrt{2}}, \quad (70)$$

$$C_{-L} = \frac{\sin \theta_1 (\cos \theta \cos \phi - i \sin \phi) - \sin \theta \cos \theta_1}{\sqrt{2}}, \quad (71)$$

and:

$$B_{LL} = \frac{q + k_s \cos \theta}{K}, \quad B_{L+} = B_{L-} = -\frac{k_s \sin \theta}{\sqrt{2}K}, \quad (72)$$

$$B_{+L} = B_{-L} = \frac{k_s \sin \theta}{\sqrt{2}K}, \quad B_{++} = B_{--} = \frac{K + q + k_s \cos \theta}{2K}, \quad (73)$$

$$B_{+-} = B_{-+} = \frac{q + k_s \cos \theta - K}{2K}. \quad (74)$$

So the nine contractions are:

$$\mathcal{E}_{LL}^L = A_{LL}B_{LL}C_{LL}, \quad \mathcal{E}_{L+}^L = A_{LL}B_{L+}C_{+L}, \quad \mathcal{E}_{L-}^L = A_{LL}B_{L-}C_{-L}, \quad (75)$$

$$\mathcal{E}_{+L}^L = A_{L+}B_{+L}C_{LL}, \quad \mathcal{E}_{++}^L = A_{L+}B_{++}C_{+L}, \quad \mathcal{E}_{+-}^L = A_{L+}B_{+-}C_{-L}, \quad (76)$$

$$\mathcal{E}_{-L}^L = A_{L-}B_{-L}C_{LL}, \quad \mathcal{E}_{-+}^L = A_{L-}B_{-+}C_{+L}, \quad \mathcal{E}_{--}^L = A_{L-}B_{--}C_{-L}. \quad (77)$$

Then we can complete the loop integral for the terms related to $D^{(L)}(k_1)$ and recognize the nonlocal signal parts:

$$\mathcal{LOOP}_{\sigma_1\sigma_2}^\lambda = \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D^{(\lambda)}(k_1, \tau_1, \tau_2)]_{ab} D^{(\sigma_1)}(K, \tau_2, \tau_3) D^{(\sigma_2)}(q, \tau_3, \tau_1) \cdot \mathcal{E}_{\sigma_1\sigma_2}^L \quad (78)$$

$$= [D^{(\lambda)}(k_1, \tau_1, \tau_2)]_{ab} \cdot \sum_{d=\pm} (\tau_1\tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \times \frac{\Gamma[-id\tilde{\nu}]^4}{(4\pi)^2} \beta^{\sigma_1 L} \beta^{\sigma_2 L} \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left(\frac{q}{2}\right)^{2id\tilde{\nu}} \left(\frac{K}{2}\right)^{2id\tilde{\nu}} \mathcal{E}_{\sigma_1\sigma_2}^L \quad (79)$$

where $\beta^{\sigma L}$ is a additional factor for the longitudinal mode:

$$\beta^{\sigma L} = \begin{cases} \frac{(-1/2+i\tilde{\nu})^2}{\tilde{\nu}^2+1/4}, & \sigma = L, \\ 1, & \sigma = \pm. \end{cases} \quad (80)$$

Since all terms proportional to $\sin \phi$ or $\cos \phi$ vanish after the azimuthal integral, we only need the ϕ -averaged angular factors. For the azimuthal average, it is important to keep the quadratic terms:

$$\overline{\sin \phi} = \overline{\cos \phi} = \overline{\sin \phi \cos \phi} = 0, \quad \overline{\sin^2 \phi} = \overline{\cos^2 \phi} = \frac{1}{2}. \quad (81)$$

We denote $z = \cos \theta$, $s_1 = \sin \theta_1$ and $c_1 = \cos \theta_1$. The non-vanishing parts are:

$$\bar{\mathcal{E}}_{LL}^L = \frac{q + k_s z}{K^2} \left[\frac{1}{2} q (1 - z^2) s_1^2 + z (qz + k_s) c_1^2 \right], \quad (82)$$

$$\bar{\mathcal{E}}_{L+}^L = \bar{\mathcal{E}}_{L-}^L = -\frac{k_s(1-z^2)}{2K^2} \left[\frac{1}{2} q z s_1^2 - (qz + k_s) c_1^2 \right], \quad (83)$$

$$\bar{\mathcal{E}}_{+L}^L = \bar{\mathcal{E}}_{-L}^L = \frac{k_s(1-z^2)}{2K^2} \left[\frac{1}{2} (qz + k_s) s_1^2 - qz c_1^2 \right], \quad (84)$$

$$\bar{\mathcal{E}}_{++}^L = \bar{\mathcal{E}}_{--}^L = \frac{K + q + k_s z}{4K^2} \left[\frac{1}{2} (z(qz + k_s) + K) s_1^2 + q(1 - z^2) c_1^2 \right], \quad (85)$$

$$\bar{\mathcal{E}}_{+-}^L = \bar{\mathcal{E}}_{-+}^L = \frac{q + k_s z - K}{4K^2} \left[\frac{1}{2} (z(qz + k_s) - K) s_1^2 + q(1 - z^2) c_1^2 \right]. \quad (86)$$

For any monomial generated by these angular factors, we first define the dimensionful integral:

$$\begin{aligned} \widetilde{\mathcal{J}}_r(m, n) &\equiv \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left(\frac{q}{2}\right)^{2id\tilde{\nu}} \left(\frac{K}{2}\right)^{2id\tilde{\nu}} q^m K^{-n} z^r \\ &= \frac{4^{-2id\tilde{\nu}} k_s^{3+4id\tilde{\nu}+m-n}}{(4\pi)^{3/2}} \mathcal{J}_r(m, n), \end{aligned} \quad (87)$$

where the dimensionless loop seeds are:

$$\mathcal{J}_r(m, n) \equiv f_r \left(x - \frac{m}{2}, x + \frac{n}{2} \right), \quad x = -id\tilde{\nu}, \quad (88)$$

and $f_1 \equiv f_c$, $f_2 \equiv f_{cc}$. The $\mathcal{J}_3$ terms below are reduced to the same loop seeds by using $z = (K^2 - q^2 - k_s^2)/(2qk_s)$:

$$\begin{aligned} \mathcal{J}_3(m, n) = & \frac{1}{8} [\mathcal{J}_0(m-3, n-6) - 3\mathcal{J}_0(m-1, n-4) - 3\mathcal{J}_0(m-3, n-4) \\ & + 3\mathcal{J}_0(m+1, n-2) + 6\mathcal{J}_0(m-1, n-2) + 3\mathcal{J}_0(m-3, n-2) \\ & - \mathcal{J}_0(m+3, n) - 3\mathcal{J}_0(m+1, n) - 3\mathcal{J}_0(m-1, n) - \mathcal{J}_0(m-3, n)]. \end{aligned} \quad (89)$$

We denote the dimensionless coefficients of the loop integrals as $\mathcal{I}_{\sigma_1\sigma_2}^\lambda(-id\tilde{\nu})$, which are defined by:

$$\mathcal{LOOP}_{\sigma_1\sigma_2}^\lambda = \left[D^{(\lambda)}(k_1, \tau_1, \tau_2) \right]_{ab} \cdot \sum_{d=\pm} (\tau_1\tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \frac{\Gamma^4(-id\tilde{\nu})}{(4\pi)^2} \tilde{\mathcal{I}}_{\sigma_1\sigma_2}^\lambda(-id\tilde{\nu}) \quad (90)$$

and explicitly:

$$\begin{aligned} \tilde{\mathcal{I}}_{\sigma_1\sigma_2}^\lambda & \equiv \beta^{\sigma_1 L} \beta^{\sigma_2 L} \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left(\frac{q}{2} \right)^{2if\tilde{\nu}} \left(\frac{K}{2} \right)^{2if\tilde{\nu}} \bar{\mathcal{E}}_{\sigma_1\sigma_2}^\lambda \\ & = \frac{4^{-2if\tilde{\nu}} k_s^{3+4if\tilde{\nu}}}{(4\pi)^{3/2}} \mathcal{I}_{\sigma_1\sigma_2}^\lambda, \quad f = \pm. \end{aligned} \quad (91)$$

Then the dimensionless coefficients of the nine loop integrals are:

$$\begin{aligned} \mathcal{I}_{LL}^L = & \frac{(-1/2 + ic\tilde{\nu})^4}{(\tilde{\nu}^2 + 1/4)^2} \cdot \left\{ \frac{s_1^2}{2} [\mathcal{J}_0(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_2(2, 2) - \mathcal{J}_3(1, 2)] \right. \\ & \left. + c_1^2 [\mathcal{J}_2(2, 2) + \mathcal{J}_3(1, 2) + \mathcal{J}_1(1, 2) + \mathcal{J}_2(0, 2)] \right\}, \end{aligned} \quad (92)$$

$$\begin{aligned} \mathcal{I}_{L+}^L = \mathcal{I}_{L-}^L = & \frac{(-1/2 + ic\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \cdot \left\{ -\frac{s_1^2}{4} [\mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)] \right. \\ & \left. + \frac{c_1^2}{2} [\mathcal{J}_1(1, 2) + \mathcal{J}_0(0, 2) - \mathcal{J}_3(1, 2) - \mathcal{J}_2(0, 2)] \right\}, \end{aligned} \quad (93)$$

$$\begin{aligned} \mathcal{I}_{+L}^L = \mathcal{I}_{-L}^L = & \frac{(-1/2 + ic\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \cdot \left\{ \frac{s_1^2}{4} [\mathcal{J}_1(1, 2) + \mathcal{J}_0(0, 2) - \mathcal{J}_3(1, 2) - \mathcal{J}_2(0, 2)] \right. \\ & \left. - \frac{c_1^2}{2} [\mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)] \right\}, \end{aligned} \quad (94)$$

$$\begin{aligned} \mathcal{I}_{++}^L = \mathcal{I}_{--}^L = & \frac{s_1^2}{8} [\mathcal{J}_2(1, 1) + \mathcal{J}_2(2, 2) + \mathcal{J}_3(1, 2) + \mathcal{J}_1(0, 1) + \mathcal{J}_1(1, 2) + \mathcal{J}_2(0, 2) \\ & + \mathcal{J}_0(0, 0) + \mathcal{J}_0(1, 1) + \mathcal{J}_1(0, 1)] \\ & + \frac{c_1^2}{4} [\mathcal{J}_0(1, 1) - \mathcal{J}_2(1, 1) + \mathcal{J}_0(2, 2) - \mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)], \end{aligned} \quad (95)$$

$$\begin{aligned} \mathcal{I}_{+-}^L = \mathcal{I}_{-+}^L = & \frac{s_1^2}{8} [\mathcal{J}_2(2, 2) + \mathcal{J}_3(1, 2) - \mathcal{J}_2(1, 1) + \mathcal{J}_1(1, 2) + \mathcal{J}_2(0, 2) - \mathcal{J}_1(0, 1) \\ & - \mathcal{J}_0(1, 1) - \mathcal{J}_1(0, 1) + \mathcal{J}_0(0, 0)] \\ & + \frac{c_1^2}{4} [\mathcal{J}_0(2, 2) - \mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2) - \mathcal{J}_0(1, 1) + \mathcal{J}_2(1, 1)]. \end{aligned} \quad (96)$$

So this part of the triangle loop integral can be written as:

$$\begin{aligned}
\mathcal{LOOP}^L &= \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{LOOP}_{\sigma_1 \sigma_2}^L \\
&= \left[D^{(L)}(k_1, \tau_1, \tau_2) \right]_{ab} \cdot \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \\
&\quad \times \frac{4^{-2id\tilde{\nu}} k_s^3 + 4id\tilde{\nu} \Gamma^4(-id\tilde{\nu})}{(4\pi)^{7/2}} \mathcal{I}^L,
\end{aligned} \tag{97}$$

where we define $\mathcal{I}^L$ is the sum of the nine contractions:

$$\mathcal{I}^L \equiv \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{I}_{\sigma_1 \sigma_2}^L = (\mathcal{I}_{LL}^L + 2\mathcal{I}_{L+}^L + 2\mathcal{I}_{+L}^L + 2\mathcal{I}_{++}^L + 2\mathcal{I}_{+-}^L) \tag{98}$$

Equivalently:

$$\mathcal{LOOP}^L = \left[D^{(L)}(k_1, \tau_1, \tau_2) \right]_{ab} \cdot \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \mathcal{B}_d^L, \tag{99}$$

with:

$$\mathcal{B}_d^L = \frac{4^{-2id\tilde{\nu}} k_s^3 + 4id\tilde{\nu} \Gamma^4(-id\tilde{\nu})}{(4\pi)^{7/2}} \mathcal{I}^{(L)}. \tag{100}$$

Now we repeat the same calculation for $\lambda = \pm$ terms. The vector contractions are:

$$\begin{aligned}
\mathcal{E}_{\sigma_1 \sigma_2}^{(\sigma)} &= e_{\mathbf{k}_1}^{(\sigma)i} e_{\mathbf{k}_1}^{(\sigma)j*} e_{\mathbf{K}}^{(\sigma_1)j} e_{\mathbf{K}}^{(\sigma_1)k*} e_{\mathbf{q}}^{(\sigma_2)k} e_{\mathbf{q}}^{(\sigma_2)i*} \\
&= (e_{\mathbf{k}_1}^{(\sigma)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}) (e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}) (e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(\sigma)}), \quad \sigma = \pm.
\end{aligned} \tag{101}$$

For the first and third factors we define:

$$\begin{aligned}
A_{+L} &= \frac{q \sin \theta \cos \phi \cos \theta_1 - iq \sin \theta \sin \phi - (q \cos \theta + k_s) \sin \theta_1}{\sqrt{2}K}, \\
A_{++} &= \frac{(q \cos \theta + k_s) \cos \theta_1 \cos \phi - iK \sin \phi \cos \theta_1 - i \sin \phi (q \cos \theta + k_s) + K \cos \phi + q \sin \theta \sin \theta_1}{2K}, \\
A_{+-} &= \frac{(q \cos \theta + k_s) \cos \theta_1 \cos \phi + iK \sin \phi \cos \theta_1 - i \sin \phi (q \cos \theta + k_s) - K \cos \phi + q \sin \theta \sin \theta_1}{2K}, \\
C_{L+} &= \frac{\sin \theta \cos \phi \cos \theta_1 + i \sin \theta \sin \phi - \cos \theta \sin \theta_1}{\sqrt{2}}, \\
C_{++} &= \frac{\cos \theta \cos \theta_1 \cos \phi + i \sin \phi \cos \theta_1 + i \cos \theta \sin \phi + \cos \phi + \sin \theta \sin \theta_1}{2}, \\
C_{-+} &= \frac{\cos \theta \cos \theta_1 \cos \phi - i \sin \phi \cos \theta_1 + i \cos \theta \sin \phi - \cos \phi + \sin \theta \sin \theta_1}{2}.
\end{aligned} \tag{102}$$

The $D^{(-)}(k_1)$ factors are obtained by complex conjugating the external polarization:

$$\begin{aligned}
A_{-L} &= \frac{q \sin \theta \cos \phi \cos \theta_1 + i q \sin \theta \sin \phi - (q \cos \theta + k_s) \sin \theta_1}{\sqrt{2}K}, \\
A_{-+} &= \frac{(q \cos \theta + k_s) \cos \theta_1 \cos \phi - i K \sin \phi \cos \theta_1 + i \sin \phi (q \cos \theta + k_s) - K \cos \phi + q \sin \theta \sin \theta_1}{2K}, \\
A_{--} &= \frac{(q \cos \theta + k_s) \cos \theta_1 \cos \phi + i K \sin \phi \cos \theta_1 + i \sin \phi (q \cos \theta + k_s) + K \cos \phi + q \sin \theta \sin \theta_1}{2K}, \\
C_{L-} &= \frac{\sin \theta \cos \phi \cos \theta_1 - i \sin \theta \sin \phi - \cos \theta \sin \theta_1}{\sqrt{2}}, \\
C_{+-} &= \frac{\cos \theta \cos \theta_1 \cos \phi + i \sin \phi \cos \theta_1 - i \cos \theta \sin \phi - \cos \phi + \sin \theta \sin \theta_1}{2}, \\
C_{--} &= \frac{\cos \theta \cos \theta_1 \cos \phi - i \sin \phi \cos \theta_1 - i \cos \theta \sin \phi + \cos \phi + \sin \theta \sin \theta_1}{2}.
\end{aligned} \tag{104}$$

The middle factor $B_{\sigma_1 \sigma_2}$ is the same as before. Therefore the nine contractions are:

$$\mathcal{E}_{LL}^\sigma = A_{\sigma L} B_{LL} C_{L\sigma}, \quad \mathcal{E}_{L+}^\sigma = A_{\sigma L} B_{L+} C_{+\sigma}, \quad \mathcal{E}_{L-}^\sigma = A_{\sigma L} B_{L-} C_{-\sigma}, \tag{105}$$

$$\mathcal{E}_{+L}^\sigma = A_{\sigma+} B_{+L} C_{L\sigma}, \quad \mathcal{E}_{++}^\sigma = A_{\sigma+} B_{++} C_{+\sigma}, \quad \mathcal{E}_{+-}^\sigma = A_{\sigma+} B_{+-} C_{-\sigma}, \tag{106}$$

$$\mathcal{E}_{-L}^\sigma = A_{\sigma-} B_{-L} C_{L\sigma}, \quad \mathcal{E}_{-+}^\sigma = A_{\sigma-} B_{-+} C_{+\sigma}, \quad \mathcal{E}_{--}^\sigma = A_{\sigma-} B_{--} C_{-\sigma}. \tag{107}$$

After the azimuthal average, we can get the non-vanishing parts of the nine contractions:

$$\begin{aligned}
\bar{\mathcal{E}}_{LL}^+ &= \frac{q + k_s z}{2K^2} \left[\frac{q(1 + c_1^2)}{2} (1 - z^2) + (qz + k_s) z s_1^2 \right] \\
\bar{\mathcal{E}}_{L+}^+ &= \left(-\frac{k_s(1 - z^2)}{4K^2} \right) \left[q c_1 + qz \frac{1 + c_1^2}{2} - (qz + k_s) s_1^2 \right] \\
\bar{\mathcal{E}}_{L-}^+ &= \left(-\frac{k_s(1 - z^2)}{4K^2} \right) \left[-q c_1 + qz \frac{1 + c_1^2}{2} - (qz + k_s) s_1^2 \right] \\
\bar{\mathcal{E}}_{+L}^+ &= \left(\frac{k_s(1 - z^2)}{4K^2} \right) \left[K c_1 + (qz + k_s) \frac{1 + c_1^2}{2} - qz s_1^2 \right] \\
\bar{\mathcal{E}}_{-L}^+ &= \left(\frac{k_s(1 - z^2)}{4K^2} \right) \left[-K c_1 + (qz + k_s) \frac{1 + c_1^2}{2} - qz s_1^2 \right] \\
\bar{\mathcal{E}}_{++}^+ &= \left(\frac{K + q + k_s z}{8K^2} \right) \left[(qz + k_s) \frac{c_1^2 z + 2c_1 + z}{2} + K \frac{c_1^2 + 2c_1 + 1}{2} + q s_1^2 (1 - z^2) \right] \\
\bar{\mathcal{E}}_{--}^+ &= \left(\frac{K + q + k_s z}{8K^2} \right) \left[(qz + k_s) \frac{c_1^2 z - 2c_1 + z}{2} + K \frac{c_1^2 - 2c_1 + 1}{2} + q s_1^2 (1 - z^2) \right] \\
\bar{\mathcal{E}}_{+-}^+ &= \left(\frac{q + k_s z - K}{8K^2} \right) \left[(qz + k_s) \frac{c_1^2 z - 2c_1 + z}{2} - K \frac{c_1^2 - 2c_1 + 1}{2} + q s_1^2 (1 - z^2) \right] \\
\bar{\mathcal{E}}_{-+}^+ &= \left(\frac{q + k_s z - K}{8K^2} \right) \left[(qz + k_s) \frac{c_1^2 z + 2c_1 + z}{2} - K \frac{c_1^2 + 2c_1 + 1}{2} + q s_1^2 (1 - z^2) \right]
\end{aligned} \tag{108}$$

Applying the same loop-seed map as above, we get the dimensionless coefficient:

$$\begin{aligned}
\mathcal{I}_{LL}^+ &= \frac{(-1/2 + id\tilde{\nu})^4}{(\tilde{\nu}^2 + 1/4)^2} \cdot \left\{ \frac{1 + c_1^2}{4} [\mathcal{J}_0(2, 2) - \mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)] \right. \\
&\quad \left. + \frac{s_1^2}{2} [\mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) + \mathcal{J}_3(1, 2) + \mathcal{J}_2(0, 2)] \right\} \\
\mathcal{I}_{L+}^+ + \mathcal{I}_{L-}^+ + \mathcal{I}_{+L}^+ + \mathcal{I}_{-L}^+ &= \frac{(-1/2 + id\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \frac{2 + s_1^2}{4} [\mathcal{J}_0(0, 2) - \mathcal{J}_2(0, 2)] \\
\mathcal{I}_{++}^+ + \mathcal{I}_{--}^+ + \mathcal{I}_{+-}^+ + \mathcal{I}_{-+}^+ &= \frac{1 + c_1^2}{4} [\mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) + \mathcal{J}_3(1, 2) + \mathcal{J}_2(0, 2) + \mathcal{J}_0(0, 0)] \\
&\quad + \frac{s_1^2}{2} [\mathcal{J}_0(2, 2) - \mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)]
\end{aligned} \tag{109}$$

So the sum of these 9 contraction:

$$\mathcal{I}^+ \equiv \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{I}_{\sigma_1 \sigma_2}^+ \tag{110}$$

Also for the $D^{(-)}(k_1)$ factors, the polarization contractions are:

$$\bar{\mathcal{E}}_{LL}^- = \bar{\mathcal{E}}_{LL}^+, \quad \bar{\mathcal{E}}_{L+}^- = \bar{\mathcal{E}}_{L-}^+, \quad \bar{\mathcal{E}}_{L-}^- = \bar{\mathcal{E}}_{L+}^+, \tag{111}$$

$$\bar{\mathcal{E}}_{+L}^- = \bar{\mathcal{E}}_{-L}^+, \quad \bar{\mathcal{E}}_{-L}^- = \bar{\mathcal{E}}_{+L}^+, \quad \bar{\mathcal{E}}_{++}^- = \bar{\mathcal{E}}_{--}^+, \tag{112}$$

$$\bar{\mathcal{E}}_{--}^- = \bar{\mathcal{E}}_{++}^+, \quad \bar{\mathcal{E}}_{+-}^- = \bar{\mathcal{E}}_{-+}^+, \quad \bar{\mathcal{E}}_{-+}^- = \bar{\mathcal{E}}_{+-}^+. \tag{113}$$

So the sum of these 9 contraction:

$$\mathcal{I}^- \equiv \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{I}_{\sigma_1 \sigma_2}^- = \mathcal{I}^+ \tag{114}$$

So the transverse part of the triangle loop integral is:

$$\mathcal{LOOP}^\sigma = \left[D^{(\sigma)}(k_1, \tau_1, \tau_2) \right]_{ab} \cdot \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \mathcal{B}_d^\sigma, \tag{115}$$

with the bubble subgraph contribution:

$$\mathcal{B}_d^\sigma = \frac{\Gamma^4(-id\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2id\tilde{\nu}} k_s^{3+4id\tilde{\nu}} \mathcal{I}^\sigma, \quad \sigma = \pm. \tag{116}$$

So the total loop integral becomes:

$$\mathcal{LOOP} = \sum_{\lambda=T, \pm} \left[D^{(\lambda)}(k_1, \tau_1, \tau_2) \right]_{ab} \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \mathcal{B}_d^\lambda \tag{117}$$

Then we also deal with the left and right subgraph contribution:

$$\begin{aligned}
\mathcal{T}_d^{(L), \lambda} &= (ig^2 h_0)^2 \sum_{a, b=\pm} (ab) \int \frac{d\tau_1}{|H\tau_1|^2} \frac{d\tau_2}{|H\tau_2|^2} G_a(k_1; \tau_1) G_b(k_2; \tau_2) \\
&\quad \times \left[D^{(\lambda)}(k_1, \tau_1, \tau_2) \right]_{ab} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}}
\end{aligned} \tag{118}$$

$$\begin{aligned}
&= (ig^2 h_0)^2 \frac{H^2}{4k_1^3 k_2^3} \sum_{a, b=\pm} (ab) \int d\tau_1 d\tau_2 (1 - iak_1 \tau_1)(1 - ibk_2 \tau_2) \\
&\quad \times \left[D^{(\lambda)}(k_1, \tau_1, \tau_2) \right]_{ab} (-\tau_1)^{-3/2+id\tilde{\nu}} (-\tau_2)^{-3/2+id\tilde{\nu}} e^{iak_1 \tau_1 + ibk_2 \tau_2}
\end{aligned} \tag{119}$$

For $\lambda = L$, this integral is given by the tree seed in [1]:

$$\mathcal{T}_d^{(L),L} = (ig^2 h_0)^2 \frac{k_{12}^{-2id\tilde{\nu}}}{4k_1^3 k_2^3} \cdot \mathcal{P}_d^{(2),L}(\tilde{\nu}) \quad (120)$$

where the coefficient is:

$$\begin{aligned} \mathcal{P}_d^{(2),L}(\tilde{\nu}) = -2^{2id\tilde{\nu}} \cdot \sum_{a,b=\pm} \left[+\tilde{\mathcal{K}}_{ab}^{-\frac{3}{2}+id\tilde{\nu},-\frac{3}{2}+id\tilde{\nu}}(1,1) - ia\tilde{\mathcal{K}}_{ab}^{-\frac{1}{2}+id\tilde{\nu},-\frac{3}{2}+id\tilde{\nu}}(1,1) \right. \\ \left. - ib\tilde{\mathcal{K}}_{ab}^{-\frac{3}{2}+id\tilde{\nu},-\frac{1}{2}+id\tilde{\nu}}(1,1) - ab\tilde{\mathcal{K}}_{ab}^{-\frac{1}{2}+id\tilde{\nu},-\frac{1}{2}+id\tilde{\nu}}(1,1) \right] \end{aligned} \quad (121)$$

and here the $\mathcal{K}_{ab}^{p_1 p_2} = \mathcal{I}_{ab}^{(L)p_1 p_2}$ in [1]. The opposite-sign seeds:

$$\begin{aligned} \mathcal{K}_{\pm\mp}^{p_1 p_2}(1,1) = \frac{e^{\mp i\pi(p_1-p_2)/2}}{4\pi(\tilde{\nu}^2 + \frac{1}{4})} \left[\mathbf{H}_{\tilde{\nu}}^{p_1}(1) \left(\frac{1}{2}\right)^{i\tilde{\nu}} + \mathbf{H}_{-\tilde{\nu}}^{p_1}(1) \left(\frac{1}{2}\right)^{-i\tilde{\nu}} \right] \\ \times \left[\mathbf{H}_{\tilde{\nu}}^{p_2}(1) \left(\frac{1}{2}\right)^{i\tilde{\nu}} + \mathbf{H}_{-\tilde{\nu}}^{p_2}(1) \left(\frac{1}{2}\right)^{-i\tilde{\nu}} \right] \end{aligned} \quad (122)$$

where we define:

$$\begin{aligned} \mathbf{H}_{\tilde{\nu}}^p(1) \equiv \left\{ \left(\frac{1}{2} - i\tilde{\nu}\right) \Gamma \left[-i\tilde{\nu}, p + \frac{3}{2} + i\tilde{\nu} \right] {}_2F_1 \left[\frac{p}{2} + \frac{3}{4} + \frac{i\tilde{\nu}}{2}, \frac{p}{2} + \frac{5}{4} + \frac{i\tilde{\nu}}{2} \middle| 1 \right] \right. \\ \left. + \frac{1}{2} \Gamma \left[-1 - i\tilde{\nu}, p + \frac{7}{2} + i\tilde{\nu} \right] {}_2F_1 \left[\frac{p}{2} + \frac{7}{4} + \frac{i\tilde{\nu}}{2}, \frac{p}{2} + \frac{9}{4} + \frac{i\tilde{\nu}}{2} \middle| 1 \right] \right\} \end{aligned} \quad (123)$$

And the same-sign seeds:

$$\mathcal{K}_{\pm\pm}^{p_1 p_2}(1,1) = \frac{-ie^{\mp i\pi(p_1+p_2)/2}}{4\pi(\tilde{\nu}^2 + \frac{1}{4})} \left[e^{\pi\tilde{\nu}} \mathbf{H}_{\tilde{\nu}}^{p_1}(1) \left(\frac{1}{2}\right)^{i\tilde{\nu}} + e^{-\pi\tilde{\nu}} \mathbf{H}_{-\tilde{\nu}}^{p_1}(1) \left(\frac{1}{2}\right)^{-i\tilde{\nu}} \right] \quad (124)$$

$$\begin{aligned} \times \left[\mathbf{H}_{\tilde{\nu}}^{p_2}(1) \left(\frac{1}{2}\right)^{i\tilde{\nu}} + \mathbf{H}_{-\tilde{\nu}}^{p_2}(1) \left(\frac{1}{2}\right)^{-i\tilde{\nu}} \right] \\ + \frac{e^{\mp i\pi(p_1+p_2)/2}}{2(\tilde{\nu}^2 + \frac{1}{4})} \sum_{n_1, n_2=0}^{\infty} \frac{(-1)^{n_{12}}}{n_1! n_2!} \left(\frac{1}{2}\right)^{2n_{12}} \end{aligned} \quad (125)$$

$$\times \frac{i}{\tilde{\nu}} \left(2n_1 - \frac{1}{2} + i\tilde{\nu} \right) \left(2n_2 - \frac{1}{2} - i\tilde{\nu} \right) (-i\tilde{\nu})_{-n_1} (i\tilde{\nu})_{-n_2} \quad (126)$$

$$\times \mathcal{F} \left[\begin{matrix} 2n_2 + p_2 + \frac{3}{2} - i\tilde{\nu}, & 2n_{12} + p_1 + p_2 + 3 \\ 2n_2 + p_2 + \frac{5}{2} - i\tilde{\nu} \end{matrix} \middle| -1 \right] + (\tilde{\nu} \rightarrow -\tilde{\nu}) \quad (127)$$

For $\lambda = \pm$, this integral can also be given by the tree seed in [2]:

$$\begin{aligned} \mathcal{T}_d^{(L),\pm} = (ig^2 h_0)^2 \frac{H^2}{4k_1^3 k_2^3} \sum_{a,b=\pm} (ab) \int d\tau_1 d\tau_2 (1 - iak_1\tau_1)(1 - ibk_2\tau_2) \\ \times \left[D^{(\text{scalar})}(k_1, \tau_1, \tau_2) \right]_{ab} (-\tau_1)^{-5/2+id\tilde{\nu}} (-\tau_2)^{-5/2+id\tilde{\nu}} e^{iak_1\tau_1 + ibk_2\tau_2} \end{aligned} \quad (128)$$

$$= (ig^2 h_0)^2 \frac{k_{12}^{-2id\tilde{\nu}}}{4k_1^3 k_2^3} \cdot \mathcal{P}_d^{(2),\pm}(\tilde{\nu}) \quad (129)$$

where the coefficient is:

$$\mathcal{P}_d^{(2),\pm}(\tilde{\nu}) = -2^{2id\tilde{\nu}} \cdot \sum_{a,b=\pm} \left[+\tilde{\mathcal{M}}_{ab}^{-\frac{5}{2}+id\tilde{\nu},-\frac{5}{2}+id\tilde{\nu}}(1,1) - ia\tilde{\mathcal{M}}_{ab}^{-\frac{3}{2}+id\tilde{\nu},-\frac{5}{2}+id\tilde{\nu}}(1,1) \right. \\ \left. -ib\tilde{\mathcal{M}}_{ab}^{-\frac{5}{2}+id\tilde{\nu},-\frac{3}{2}+id\tilde{\nu}}(1,1) - ab\tilde{\mathcal{M}}_{ab}^{-\frac{3}{2}+id\tilde{\nu},-\frac{3}{2}+id\tilde{\nu}}(1,1) \right] \quad (130)$$

and here $\mathcal{M}_{ab}^{p_1 p_2} = \mathcal{I}_{ab}^{p_1 p_2}$ in [2]. The opposite-sign seeds:

$$\mathcal{M}_{\pm\mp}^{p_1 p_2}(1,1) = \frac{e^{\mp i p_{12} \pi/2}}{2^{5+p_{12}}} \Gamma \left[\begin{matrix} \frac{5}{2} + p_1 - i\tilde{\nu}, & \frac{5}{2} + p_1 + i\tilde{\nu}, & \frac{5}{2} + p_2 - i\tilde{\nu}, & \frac{5}{2} + p_2 + i\tilde{\nu} \\ 3 + p_1, & 3 + p_2 \end{matrix} \right] \quad (131)$$

And the same-sign seeds:

$$\mathcal{M}_{\pm\pm}^{p_1 p_2}(1,1) = \frac{\pm i e^{\mp i p_{12} \pi/2} e^{-\pi\tilde{\nu}}}{2^{5+p_{12}}} \Gamma \left[\begin{matrix} \frac{5}{2} + p_1 - i\tilde{\nu}, & \frac{5}{2} + p_1 + i\tilde{\nu}, & \frac{5}{2} + p_2 - i\tilde{\nu}, & \frac{5}{2} + p_2 + i\tilde{\nu} \\ 3 + p_1, & 3 + p_2 \end{matrix} \right] \\ - \frac{e^{\mp i p_{12} \pi/2}}{2^{5+p_{12}}} \Gamma \left[5 + p_{12}, \quad \frac{5}{2} + p_1 \pm i\tilde{\nu}, \quad \frac{5}{2} + p_2 \pm i\tilde{\nu} \right] \\ \cdot {}_3\tilde{F}_2 \left[\begin{matrix} 5 + p_{12}, & \frac{1}{2} \pm i\tilde{\nu}, & 1 \\ \frac{7}{2} + p_1 \pm i\tilde{\nu}, & \frac{7}{2} + p_2 \pm i\tilde{\nu} & 1 \end{matrix} \right]. \quad (132)$$

And for the right subgraph:

$$\mathcal{T}_d^{(R)} = (ig^2) \sum_c c \int \frac{d\tau_3}{|H\tau_3|^2} G_c(k_3; \tau_1) G_c(k_4; \tau_3) (-\tau_3)^{1+2id\tilde{\nu}} \\ = (ig^2) \frac{H^2}{4k_3^3 k_4^3} \sum_{c=\pm} c \int_{-\infty}^0 d\tau_3 (-\tau_3)^{-1+2ic\tilde{\nu}} (1 - ick_{34}\tau_3 - k_3 k_4 \tau_3^2) e^{ick_{34}\tau_3} \\ = (ig^2) \frac{H^2}{4k_3^3 k_4^3} \times k_{34}^{-2id\tilde{\nu}} \cdot \mathcal{P}_d^{(1)}(\tilde{\nu}) \quad (133)$$

where the coefficient is:

$$\mathcal{P}_d^{(1)}(\tilde{\nu}) = \frac{2 + id\tilde{\nu}}{2id\tilde{\nu}} \sinh(\pi d\tilde{\nu}) \Gamma(2 + 2id\tilde{\nu}) \quad (134)$$

So one of the triangle diagram:

$$\langle \delta h^4 \rangle^{\text{Tri1}} = \sum_{d=\pm} \sum_{\lambda=L,\pm} \mathcal{T}_d^{(L),\lambda} \cdot \mathcal{B}_d^\lambda \cdot \mathcal{T}_d^{(R)} \quad (135)$$

$$= (ig^2)(ig^2 h_0)^2 \frac{H^2 k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times C^{\text{Tri}}(\tilde{\nu}, \theta_1) + \text{c.c.} \quad (136)$$

where the magnitude is:

$$C^{\text{Tri}}(\tilde{\nu}, \theta_1) = \sum_{\lambda=L,\pm} [\mathcal{P}_+^{(1)}(\tilde{\nu}) \mathcal{P}_+^{(2),\lambda}(\tilde{\nu})] \cdot \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \cdot \mathcal{I}^\lambda(\tilde{\nu}, \theta_1)|_{d=+} \quad (137)$$

Also for another triangle:

$$\langle \delta h^4 \rangle^{\text{Tri2}} = \langle \delta h^4 \rangle^{\text{Tri1}}|_{k_1 \longleftrightarrow k_3, k_2 \longleftrightarrow k_4, \theta_1 \longleftrightarrow \theta_3}, \quad (138)$$

So the total triangle diagram contributions:

$$\langle \delta h^4 \rangle^{\text{Tri}} = \langle \delta h^4 \rangle^{\text{Tri1}} + \langle \delta h^4 \rangle^{\text{Tri2}} \\ = (ig^2)(ig^2 h_0)^2 \frac{H^2 k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times (C^{\text{Tri}}(\tilde{\nu}, \theta_1) + C^{\text{Tri}}(\tilde{\nu}, \theta_3)) + \text{c.c.} \quad (139)$$

4 Box Diagram

For the box diagrams:

$$\begin{aligned}
\langle \delta h^4 \rangle^{\text{Box}} &= (ig^2 h_0)^4 \sum_{a,b,c,d=\pm} (abcd) \int \frac{d\tau_1}{|H\tau_1|^2} \frac{d\tau_2}{|H\tau_2|^2} \frac{d\tau_3}{|H\tau_3|^2} \frac{d\tau_4}{|H\tau_4|^2} \\
&\times G_a(k_1; \tau_1) G_a(k_2; \tau_2) G_c(k_3; \tau_3) G_d(k_4; \tau_4) \\
&\times \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)]_{bc} \\
&\times [D_{kl}(\mathbf{q} + \mathbf{k}_3, \tau_3, \tau_4)]_{cd} [D_{li}(\mathbf{q}, \tau_4, \tau_1)]_{da}
\end{aligned} \tag{140}$$

Also we deal with the loop integral:

$$\begin{aligned}
\mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)]_{bc} \\
&\times [D_{kl}(\mathbf{q} + \mathbf{k}_3, \tau_3, \tau_4)]_{cd} [D_{li}(\mathbf{q}, \tau_4, \tau_1)]_{da}
\end{aligned} \tag{141}$$

Using the cutting rule:

$$\begin{aligned}
\mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)] \\
&\times [D_{kl}(\mathbf{k}_3, \tau_3, \tau_4)]_{cd} [D_{li}(\mathbf{q}, \tau_4, \tau_1)]
\end{aligned} \tag{142}$$

Expanding the four propagators in the helicity basis, the loop integral becomes:

$$\begin{aligned}
\mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(k_1, \tau_1, \tau_2) e_{\mathbf{k}_1}^{(L)i} e_{\mathbf{k}_1}^{(L)j*} + D^{(\pm)}(k_1, \tau_1, \tau_2) e_{\mathbf{k}_1}^{(\pm)i} e_{\mathbf{k}_1}^{(\pm)j*} \right]_{ab} \\
&\times \left[D^{(L)}(K, \tau_2, \tau_3) e_{\mathbf{K}}^{(L)j} e_{\mathbf{K}}^{(L)k*} + D^{(\pm)}(K, \tau_2, \tau_3) e_{\mathbf{K}}^{(\pm)j} e_{\mathbf{K}}^{(\pm)k*} \right] \\
&\times \left[D^{(L)}(k_3, \tau_3, \tau_4) e_{\mathbf{k}_3}^{(L)k} e_{\mathbf{k}_3}^{(L)l*} + D^{(\pm)}(k_3, \tau_3, \tau_4) e_{\mathbf{k}_3}^{(\pm)k} e_{\mathbf{k}_3}^{(\pm)l*} \right]_{cd} \\
&\times \left[D^{(L)}(q, \tau_4, \tau_1) e_{\mathbf{q}}^{(L)l} e_{\mathbf{q}}^{(L)i*} + D^{(\pm)}(q, \tau_4, \tau_1) e_{\mathbf{q}}^{(\pm)l} e_{\mathbf{q}}^{(\pm)i*} \right],
\end{aligned} \tag{143}$$

where $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, and each repeated $\pm$ label is summed over the two transverse helicities.

To organize the nine external-helicity combinations, we choose the same convention for $\mathbf{k}_1$ as in the triangle diagram and set

$$\mathbf{k}_1 = k_1(\sin \theta_1, 0, \cos \theta_1), \tag{144}$$

$$\mathbf{k}_3 = k_3(\sin \theta_3 \cos \phi_3, \sin \theta_3 \sin \phi_3, \cos \theta_3). \tag{145}$$

Let $\lambda_1, \lambda_2 = L, \pm$ label the polarizations of the hard propagators with momenta $\mathbf{k}_1$ and $\mathbf{k}_3$, respectively, and let $\sigma_1, \sigma_2 = L, \pm$ label the two soft propagators. Keeping the latter two helicities explicit, as in the Triangle calculation, define

$$\begin{aligned}
\mathcal{E}_{\sigma_1 \sigma_2}^{\lambda_1 \lambda_2} &\equiv e_{\mathbf{k}_1}^{(\lambda_1)i} e_{\mathbf{k}_1}^{(\lambda_1)j*} e_{\mathbf{K}}^{(\sigma_1)j} e_{\mathbf{K}}^{(\sigma_1)k*} e_{\mathbf{k}_3}^{(\lambda_2)k} e_{\mathbf{k}_3}^{(\lambda_2)l*} e_{\mathbf{q}}^{(\sigma_2)l} e_{\mathbf{q}}^{(\sigma_2)i*} \\
&= (e_{\mathbf{k}_1}^{(\lambda_1)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}) (e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{k}_3}^{(\lambda_2)}) (e_{\mathbf{k}_3}^{(\lambda_2)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}) (e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(\lambda_1)}).
\end{aligned} \tag{146}$$

Their explicit contraction check is collected in Appendix B. The dimensionful loop integral is defined by

$$\begin{aligned}\tilde{\mathcal{I}}_{\sigma_1\sigma_2}^{\lambda_1\lambda_2} &\equiv \beta^{\sigma_1 L} \beta^{\sigma_2 L} \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left(\frac{q}{2}\right)^{2if\tilde{\nu}} \left(\frac{K}{2}\right)^{2if\tilde{\nu}} \bar{\mathcal{E}}_{\sigma_1\sigma_2}^{\lambda_1\lambda_2} \\ &= \frac{4^{-2if\tilde{\nu}} k_s^{3+4if\tilde{\nu}}}{(4\pi)^{3/2}} \mathcal{I}_{\sigma_1\sigma_2}^{\lambda_1\lambda_2}(\theta_1, \theta_3, \phi_3), \quad f = \pm.\end{aligned}\quad (147)$$

Here the index f labels the two nonlocal branches and is distinct from the SK index d of the box diagram. The loop integral can then be written in the same form as the triangle result:

$$\begin{aligned}\mathcal{LOOP} &= \sum_{\lambda_1, \lambda_2=L, \pm} \left[D^{(\lambda_1)}(k_1, \tau_1, \tau_2) \right]_{ab} \left[D^{(\lambda_2)}(k_3, \tau_3, \tau_4) \right]_{cd} \\ &\quad \times \sum_{f=\pm} (\tau_1 \tau_2 \tau_3 \tau_4)^{1/2+if\tilde{\nu}} \mathcal{B}_f^{\lambda_1\lambda_2}(k_1, k_3),\end{aligned}\quad (148)$$

where

$$\begin{aligned}\mathcal{B}_f^{\lambda_1\lambda_2}(k_1, k_3) &= \frac{\Gamma^4(-if\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2if\tilde{\nu}} k_s^{3+4if\tilde{\nu}} \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{I}_{\sigma_1\sigma_2}^{\lambda_1\lambda_2}(\theta_1, \theta_3, \phi_3) \\ &= \frac{\Gamma^4(-if\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2if\tilde{\nu}} k_s^{3+4if\tilde{\nu}} \mathcal{I}^{\lambda_1\lambda_2}(\theta_1, \theta_3, \phi_3)\end{aligned}\quad (149)$$

Then we can deal with the left and right subgraph, which is identical to the left subgraph contribution of triangle diagrams:

$$\mathcal{T}_f^{(L), \lambda_1} = \left(ig^2 h_0\right)^2 \frac{k_{12}^{-2id\tilde{\nu}}}{4k_1^3 k_2^3} \cdot \mathcal{P}_f^{(2), \lambda_1}(\tilde{\nu}) \quad (150)$$

$$\mathcal{T}_f^{(R), \lambda_2} = \left(ig^2 h_0\right)^2 \frac{k_{34}^{-2id\tilde{\nu}}}{4k_3^3 k_4^3} \cdot \mathcal{P}_f^{(2), \lambda_2}(\tilde{\nu}) \quad (151)$$

So the box diagram:

$$\langle \delta h^4 \rangle^{\text{Box}} = \sum_{f=\pm} \sum_{\lambda_1, \lambda_2=L, \pm} \mathcal{T}_f^{(L), \lambda_1} \cdot \mathcal{B}_f^{\lambda_1\lambda_2} \cdot \mathcal{T}_f^{(R), \lambda_2} \quad (152)$$

$$= (ig^2 h_0)^4 \frac{H^2 k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times C^{\text{Box}}(\tilde{\nu}, \theta_1) + \text{c.c.} \quad (153)$$

where the magnitude is:

$$C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = \sum_{\lambda_z, \lambda_2=L, \pm} [\mathcal{P}_+^{(2), \lambda_1}(\tilde{\nu}) \mathcal{P}_+^{(2), \lambda_2}(\tilde{\nu})] \cdot \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \cdot \mathcal{I}^{\lambda_1\lambda_2}(\theta_1, \theta_3, \phi_3)|_{f=+} \quad (154)$$

A Check of the Box Polarization Contractions

For the Box diagram, introduce the shorthand

$$\hat{q} = (u \cos \phi, u \sin \phi, z), \quad \hat{K} = \frac{1}{K}(qu \cos \phi, qu \sin \phi, qz + k_s), \quad (155)$$

$$\hat{k}_1 = (s_1, 0, c_1), \quad \hat{k}_3 = (s_3 \cos \phi_3, s_3 \sin \phi_3, c_3), \quad (156)$$

where $s_3 = \sin \theta_3$ and $c_3 = \cos \theta_3$. For a unit vector with polar angles (θ_p, ϕ_p) , we use

$$e_{\mathbf{p}}^{(L)} = (\sin \theta_p \cos \phi_p, \sin \theta_p \sin \phi_p, \cos \theta_p), \quad (157)$$

$$e_{\mathbf{p}}^{(\sigma)} = \frac{1}{\sqrt{2}}(\cos \theta_p \cos \phi_p - \sigma i \sin \phi_p, \cos \theta_p \sin \phi_p + \sigma i \cos \phi_p, -\sin \theta_p), \quad \sigma = \pm 1. \quad (158)$$

Thus $e_{\mathbf{k}_1}^{(\lambda_1)}$ and $e_{\mathbf{k}_3}^{(\lambda_2)}$ follow by taking $(\theta_p, \phi_p) = (\theta_1, 0)$ and (θ_3, ϕ_3) , respectively.

For the physical Box integral, the soft helicities must be kept explicit. Define

$$A_{\sigma_1}^{\lambda_1} \equiv e_{\mathbf{k}_1}^{(\lambda_1)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}, \quad B_{\sigma_1}^{\lambda_2} \equiv e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{k}_3}^{(\lambda_2)}, \quad (159)$$

$$C_{\sigma_2}^{\lambda_2} \equiv e_{\mathbf{k}_3}^{(\lambda_2)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}, \quad D_{\sigma_2}^{\lambda_1} \equiv e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(\lambda_1)}. \quad (160)$$

The Box angular factor is therefore

$$\mathcal{E}_{\sigma_1 \sigma_2}^{\lambda_1 \lambda_2} = A_{\sigma_1}^{\lambda_1} B_{\sigma_1}^{\lambda_2} C_{\sigma_2}^{\lambda_2} D_{\sigma_2}^{\lambda_1}, \quad (161)$$

which is the direct Box analogue of the three-factor Triangle contraction. One must first form this product, then perform the ϕ average, and only afterwards apply the loop-seed map separately for every (σ_1, σ_2) .

B Closed Gamma-function forms for $\mathcal{J}_3$ and $\mathcal{J}_4$

For a general loop seed, write $\mathcal{J}_r(m, n) = f_r(a, b)$ with

$$a = x - \frac{m}{2}, \quad b = x + \frac{n}{2}, \quad x = -if\tilde{\nu}. \quad (162)$$

The identity $z = (K^2 - q^2 - k_s^2)/(2qk_s)$ gives the following finite Gamma-function closed forms:

$$\begin{aligned} f_3(a, b) = \frac{1}{8} & \left[f_0\left(a + \frac{3}{2}, b - 3\right) - 3f_0\left(a + \frac{3}{2}, b - 2\right) - 3f_0\left(a + \frac{1}{2}, b - 2\right) \right. \\ & + 3f_0\left(a + \frac{3}{2}, b - 1\right) + 6f_0\left(a + \frac{1}{2}, b - 1\right) + 3f_0\left(a - \frac{1}{2}, b - 1\right) \\ & \left. - f_0\left(a + \frac{3}{2}, b\right) - 3f_0\left(a + \frac{1}{2}, b\right) - 3f_0\left(a - \frac{1}{2}, b\right) - f_0\left(a - \frac{3}{2}, b\right) \right], \end{aligned} \quad (163)$$

$$\begin{aligned} f_4(a, b) = \frac{1}{16} & \left[f_0(a + 2, b - 4) - 4f_0(a + 2, b - 3) - 4f_0(a + 1, b - 3) \right. \\ & + 6f_0(a + 2, b - 2) + 12f_0(a + 1, b - 2) + 6f_0(a, b - 2) \\ & - 4f_0(a + 2, b - 1) - 12f_0(a + 1, b - 1) - 12f_0(a, b - 1) - 4f_0(a - 1, b - 1) \\ & \left. + f_0(a + 2, b) + 4f_0(a + 1, b) + 6f_0(a, b) + 4f_0(a - 1, b) + f_0(a - 2, b) \right]. \end{aligned} \quad (164)$$

Here f_0 is the Gamma ratio defined in the loop-seed section. Consequently, these equations are closed expressions in Gamma functions, not new independent seeds. More generally,

$$f_r(a, b) = \frac{1}{2^r} \sum_{p+q+t=r} \frac{r!}{p!q!t!} (-1)^{q+t} f_0\left(a + \frac{r}{2} - q, b - p\right), \quad (165)$$

which provides a direct check of both formulas.

References

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